

Liver Function Test



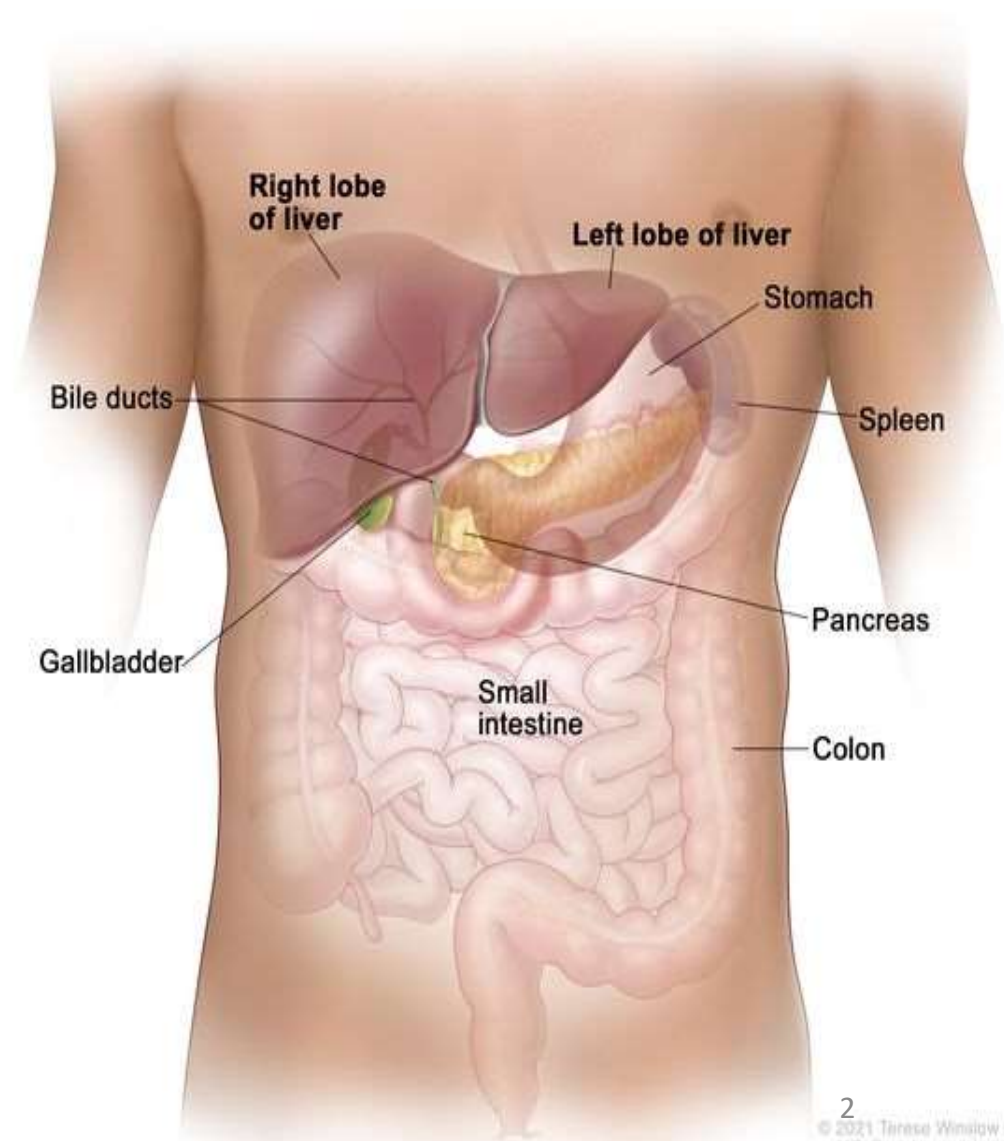
Prepared by

Dr. Fargeen E. ALmukhtar

M. Sc. Ph. D. in Clinical Biochemistry
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Introduction:

- The liver is a large organ found in the upper right quadrant of the abdomen,
- shaped like a cone, and it has a dark reddish-brown color that weighs about 1.5 Kg.
- It's an essential organ of the body that performs over 500 vital functions.



Main Functions of the Liver:

- 1. Digestion:** produces bile, which is essential for dietary fat digestion and absorption.
- 2. Metabolism:** regulates metabolism of carbs, lipids and proteins.
- 3. Detoxification:** converts harm substances into harmless that can be excreted from the body.
- 4. Storage:** stores glycogen, vitamins (A, D, E, K, B12), and minerals (iron and copper).
- 5. Synthesis:** produces essential proteins and clotting factors, carbohydrates, lipids, bile salts, vitamins, and hormones.

- 6. Immune Function:** filters bacteria and other pathogens from the bloodstream.
- 7. Hormone Regulation:** activating and inactivating of certain hormones, thus maintaining hormonal balance.
- 8. Waste Removal:** removes wastes by converting harmful substances into substances that can be excreted through bile or urine.
- 9. Energy Regulation:** by storing; releasing; and producing glucose, converting fats and proteins into energy when needed.

What are liver function tests?

- Liver function tests (LFTs) or liver chemistries, are a panel of blood tests that measure **enzymes**, **proteins**, and **other substances** to assess liver health and function.
- They help diagnose liver diseases, monitor their severity, and check for liver damage caused by different conditions.

The most common liver tests include:

1- Liver enzymes test: include alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase (ALP), and gamma glutamyl transferase (GGT). These are elevated when there's **liver injury**.

i- ALT:

- Alanine aminotransferase is more specific for liver injury than AST.
- Found mainly in the liver.
- Normal range: (5-45) IU/L
- Very high levels are most often caused by viral hepatitis, ischemic hepatitis, or injury from drugs or other chemicals.

ii- AST:

- Aspartate aminotransferase is less liver specific than ALT.
- Found mostly in the **liver**, but also in the **heart**, **kidneys**, **skeletal muscle**, and **brain**.
- Normal range: **(5-45) IU/L**
- Elevated levels can indicate liver damage or injury (hepatitis, cirrhosis), MI, muscular dystrophy, kidney disease.

Note:

AST > ALT (alcoholic liver disease)

ALT > AST (viral or non- alcoholic liver disease)

iii- ALP:

- Alkaline phosphatase is a non-specific marker of liver disease.
- Found mainly in the liver, bones, bile ducts, and placenta.
- Normal range: (30-120) IU/L
- High levels of ALP may indicate liver inflammation, blockage of the bile ducts, and bone diseases.

IV- GGT:

- γ -glutamyltransferase used for glutathione synthesis.
- An enzyme that is mainly made in the liver, kidney, and biliary tract.
- Normal range: (5-50)IU/L
- Elevated in liver and bile duct disorders, chronic alcohol consumption.

2- Total protein test: measures the total amount of protein in the blood, which includes **albumin** and **globulins**. These proteins are mainly made in the liver.

i- Albumin:

- The most abundant protein synthesized by the **liver**.
- Normal range: **(3.5 to 5) g/dL**
- It's level **decrease** in all **chronic liver diseases** like (cirrhosis, hepatitis, or liver cancer).
- High levels of albumin **rarely** due to liver problems.

ii- Globulin:

- The globulins are a family of globular proteins including (Gamma globulins known as immunoglobulins, Alpha and beta globulins, and blood clotting globulins).
- Made by the **liver** and **immune system**.
- Normal range: **(2.5 to 3.5g/dL)**
- High serum γ -globulins are observed in: **chronic liver diseases** (hepatitis and cirrhosis), **infection**, and **multiple myeloma**.

3- Bilirubin test:

- A byproduct of red blood cell breakdown.
- It is the yellowish pigment observed in jaundice.
- Normal range of Tb: 0.2 to 1.2 mg/dl
- Unconjugated/free/indirect (bilirubin-albumin complex):
0.2 to 0.8 mg/dL
- Conjugated/direct: 0.0 to 0.4 mg/dL
- High bilirubin levels are observed in: jaundice that mainly resulted from (gallstones, and acute or chronic hepatitis).

4- Prothrombin time (PT) / International Normalized Ratio (INR):

- Evaluate the extrinsic pathway of coagulation, which depends on clotting factors (II, V, VII, and X) all of which are made in the liver and required vitamin K.
- PT: measures how long it takes for a sample of the blood to clot (in seconds).
- INR: a standardized way of reporting PT (so results comparable across labs).
- Normal rang: PT (11-13.5) seconds, INR $\approx$ 1.0

➤ Prolonged PT/INR

- found in liver diseases (hepatitis, cirrhosis, acute liver failure).
- vitamin K deficiency
- clotting factor deficiency
- Warfarin therapy.

How PT/INR is measured practically?

1. Prepare your hands and the device.
2. Match the test strip code.
3. Prepare the finger
4. Perform the finger-stick
5. Apply the blood to the test strip.
6. Wait for the result.
7. Review the result.



What are they (LFTs) used for?

- To **diagnose** and **monitor** liver diseases.
- To **evaluate liver function** in patients taking medications that can affect the liver.
- To **investigate symptoms** like jaundice (yellow skin/eyes), abdominal pain, or unexplained fatigue.
- To **check for liver damage** from toxins or excessive alcohol consumption.

Case Reports

Case (1): A 58-year-old man presents with jaundice, pruritus, pale stools, and dark urine. Ultrasound shows a gallstone in the common bile duct.

Lab results:

AST: 98 IU/L

ALT: 110 IU/L

ALP: 850 IU/L

Total bilirubin: 13 mg/dl (mainly conjugated)

Albumin: 4.1 g/dl

PT/INR: 1.3

Interpretation: Very high ALP and bilirubin with only mild AST/ALT rise indicate (Obstructive cholestatic jaundice)

Case (2): A 30-year-old woman presents with confusion, jaundice, and a history of paracetamol overdose.

Lab results:

AST: 2300 IU/L

ALT: 2550 IU/L

ALP: 135 IU/L

Total bilirubin: 8.5 mg/dl (mainly conjugated)

Albumin: 3.7 g/dl

PT/INR: 4

Interpretation: Extremely high AST/ALT with markedly prolonged INR and encephalopathy indicates (acute fulminant hepatic failure).

References:

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Thank You!

